

Multi-Signal Slip Detection for a Low-Cost Robotic Gripper

Design and experimental plan for the slip-detection stage of an adaptive grasping system

Anish Talla

Academy of Engineering and Technology, Academies of Loudoun

Senior Research, AET Engineering

Interim design report, September 2026

Status. This is a design report. The gripper has not been built and no slip trials have been run. Numbers in the text are design targets, component specifications, or results from published work with the source named. Figure 2 is an illustration of the signal shape the design expects and does not contain measured data.

Abstract

A gripper holding an unfamiliar object does not know how hard to squeeze. Too little force and the object slides out; too much and a deformable object is crushed. Most published slip detectors use optical tactile sensors or dense tactile arrays, which cost far more than a high school research budget allows. This report describes the design and test plan for a slip detector built from two inexpensive sensors on one parallel-jaw gripper. Force-sensitive resistors (FSRs) under the jaw pads track the normal load at the contact, which shifts as an object starts to move. Piezoelectric discs on the finger bodies pick up the frictional micro-vibration that comes before visible motion. An Arduino Mega logs the four analog channels at a target rate of about 1000 Hz each. Slip onset is labeled independently from 240 fps phone video, and a single LED flash visible in both records aligns the video with the serial log. The experiment crosses three sensing conditions (force only, vibration only, fused) with two detectors (a rule-based threshold and a random forest) and three window lengths (25, 50, and 100 ms), scored with five-fold cross-validation grouped by trial. The planned dataset is 45 pilot trials, 90 slip trials, and 30 no-slip trials over nine combinations of surface sleeve and object mass. One constraint shapes most of the design: a detector cannot answer until its window has filled, so window length is subtracted directly from a 200 ms response budget. The intended contribution is a measurement of how much detection performance this low-cost sensor pair recovers compared with published results from instruments that cost roughly a hundred times more.

1 Introduction

1.1 Problem

If an object's mass, friction, and stiffness are known, the grip force it needs can be computed once and commanded open loop. Outside a factory those properties usually are not known, so the controller has to find the right force from feedback while it holds the object. People do this constantly and mostly without noticing. Fast-adapting mechanoreceptors in the skin respond to the small vibrations produced when skin starts to slide over a surface, and grip force rises before anything falls. A robot gripper without that feedback has two options, neither good: grip hard and damage soft objects, or grip lightly and drop things.

This project targets the interval between those two outcomes. Before an object visibly moves, part of the contact patch is already sliding while the rest is still stuck. This state is called incipient slip. A correction made here is cheap, since the object has not accelerated yet and a small increase in normal force is enough to stop it. A detector that catches incipient slip is useful to a closed-loop gripper in a way that a detector which reports the drop afterward is not.

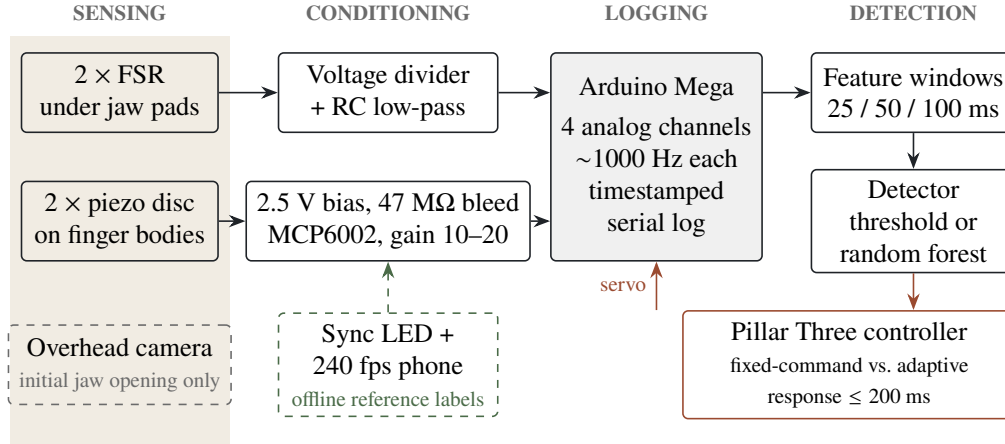


Figure 1. System architecture. Two FSRs and two piezo discs feed four analog channels on an Arduino Mega, which writes a timestamped serial log. Windowing and detection run downstream of the log. The camera (grey, dashed) only sets the initial jaw opening. The LED and video path (green, dashed) produces reference labels offline and is not part of the real-time loop.

1.2 Scope

The full project has three parts. Pillar One uses vision to choose an initial grasp. Pillar Two, covered here, detects slip from tactile signals. Pillar Three adjusts grip force in real time when slip is detected. The parts run in sequence on the robot but can be studied separately, and this report keeps the other two out of the analysis as far as possible.

Vision has one job in this study. An overhead camera measures object width and sets the initial jaw opening so that every trial starts from similar geometry. It is not an experimental variable and no vision results are reported. Recovery enters only as the consumer of the detector’s output and as the source of the response-time budget. The recovery experiment itself (140 trials comparing a fixed-command controller with an adaptive one) belongs to Pillar Three and is mentioned only where it places a requirement on the detector.

1.3 Research questions

- RQ1. Does fusing force and vibration data improve slip detection over either signal alone, measured by precision, recall, and detection delay? If so, by how much?
- RQ2. How does window length trade against detection quality, given that the window length is also a lower bound on response time?
- RQ3. Does a random forest outperform a rule-based threshold detector on this hardware? If the threshold detector is close, it is the better choice, since it is cheaper and runs on the microcontroller.
- RQ4. How does detection performance change with surface friction and object mass, and which of the two accounts for more of the variation?

1.4 Current status

The literature review and design are finished. This includes six annotated bibliographies, two of them on experimental method, a fixed hardware list, the signal-conditioning circuit, the trial matrix, and the labeling procedure. No hardware has been built. Section 7 lists the build order and milestones.

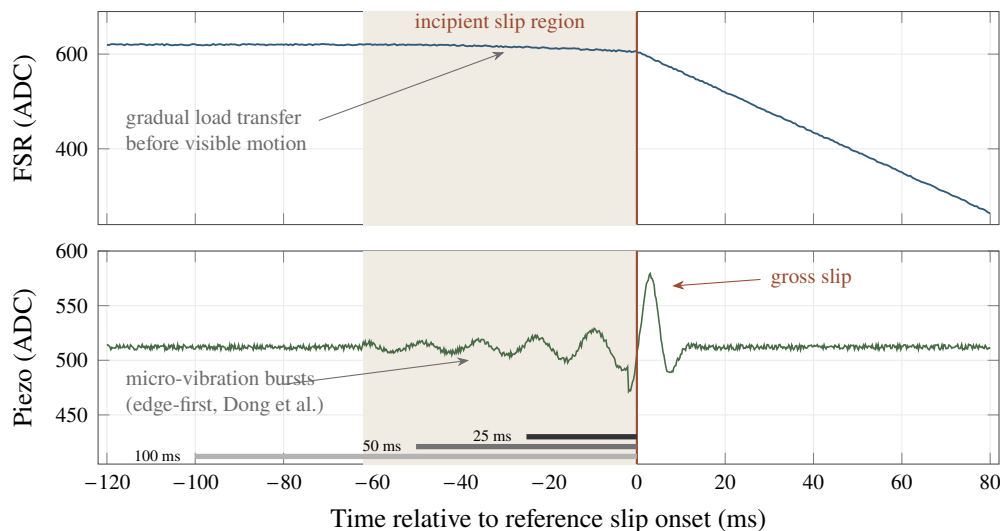


Figure 2. Illustration of expected signal behavior around a slip event (synthetic, not measured). The FSR shows slow load transfer through the incipient region before any position change is measurable, then a steep fall at gross slip. The piezo shows small frictional bursts through the same region, then a large transient. The bars at the bottom show how far back into the pre-onset interval each window length reaches.

2 Background

2.1 Incipient slip and gross slip

Gross slip is the whole object moving. It is easy to detect and it is late: by the time it can be measured, the object is already accelerating under gravity. Incipient slip comes first. Pressure across a fingertip contact is not uniform, so the shear needed to break static friction is not reached everywhere at the same moment. The edges of the contact patch, which carry less normal pressure, start to slide while the center stays stuck. As the load increases the sliding region grows inward until the entire patch moves, and the object is then in gross slip (Dong et al., 2017).

The two events look different to a sensor. Incipient slip produces weak high-frequency frictional vibration and a slow shift in normal load. Gross slip produces a large transient that is hard to miss. Detecting gross slip is easy but too late to help; detecting incipient slip means working with a much weaker signal, and that is where most of the difficulty in this project lies. Figure 2 sketches the expected shape of both signals.

2.2 Why use two sensors

FSRs and piezo discs are weak in opposite places, and that is the reason to carry both.

An FSR is a polymer film whose resistance drops as normal pressure rises. It measures slowly changing load well and fast transients badly, because the polymer responds slowly and the readout is usually low-pass filtered to control noise. During incipient slip an FSR does not register the vibration. What it registers is load transfer: the normal force at the contact falls as shear takes over and the object’s weight shifts.

A piezo disc produces charge in proportion to the rate of strain. It has essentially no response to constant load and good sensitivity to exactly the frictional vibration an FSR misses. On its own it cannot report how hard the gripper is squeezing, and that is the most important state variable for a recovery controller.

Neither sensor is enough by itself, but together they cover both halves of the event. The working hypoth-

Table 1. Hardware for Pillar Two, listed with the specification that drove each choice.

Component	Role	Relevant specification
FSR (2)	Normal load at each jaw pad	Slow response; read through a voltage divider and RC low-pass
Piezo disc (2)	Frictional vibration, one per finger body	Charge output, no DC response; needs bias and a high-impedance front end
MCP6002 op-amp	Buffer and gain for the piezo channels	Gain 10–20, output held within 0–5 V
Arduino Mega	Four-channel acquisition and serial logging	Target ~1000 Hz per channel, to be measured
MG996R servo	Drives a 3D-printed parallel-jaw linkage	Regulated 6 V, 2–3 A rail, common ground with the Arduino
Phone camera	Reference labels for slip onset	240 fps, perpendicular to the fall direction, scale reference in frame
Sync LED	Shared time zero for video and log	Switch-on appears in both records

esis is that a classifier given both signals will outperform one given either signal alone. This is tested directly as the primary comparison (RQ1) and is not assumed.

2.3 Cost

Most published slip-detection results come from hardware a school project cannot buy. GelSight-style optical sensors image a gel membrane with an internal camera and resolve contact geometry at a density resistive sensing cannot match (Dong et al., 2017; Li et al., 2018). Structured tactile arrays and multi-axis load cells sit in a similar price range. These systems work very well, but they say little about what a gripper with about \$200 of parts can do.

The gap this project addresses is a measurement gap. My literature search did not find a study that pairs FSRs with piezoelectric vibration sensing on the same low-cost gripper and quantifies how that pairing compares with the expensive instruments. Section 6.1 states the claim more precisely, along with what would disprove it.

3 Related work and its effect on the design

The sources below each changed a specific design decision, and the decision is given with the finding. Table 2 summarizes them.

3.1 Romano et al. (2011): structure of a tactile grasp controller

Romano et al. (2011) built a grasp controller for the PR2 robot that picks up, holds, and sets down unknown objects without dropping or crushing them. It uses pressure arrays on both fingertips and an accelerometer in the gripper, and moves through six states: Close, Load, Lift and Hold, Replace, Unload, and Open. The design is modeled on human grasping, with each derived tactile signal standing in for a class of human mechanoreceptor.

Two details from this paper carried over. The first is the difference in sampling rates on one robot: 24.4 Hz for the pressure arrays and 3 kHz for the accelerometer. Load and vibration have different bandwidths, which

is the same reasoning behind putting an FSR and a piezo disc on one gripper here. It also means that if vibration carries useful information at kilohertz rates, a 1000 Hz piezo channel is a compromise that has to be checked by measurement.

The second is the recovery strategy. Romano’s controller responds to slip in proportion to the event instead of by a fixed step. That distinction becomes the fixed-command versus adaptive comparison in Pillar Three, and it is why the detector here should output more than a binary flag where it can.

Romano’s controller also uses no vision; it starts from contact. This supports the division of labor in this project, where vision sets the initial jaw opening and then hands control to the tactile loop.

3.2 Van Wyk and Falco (2018): window length and sampling rate

This NIST study (Van Wyk and Falco, 2018) evaluated three commercial tactile sensors as slip detectors while varying window size, sampling rate, material, slip speed, and unit-to-unit variation. LSTM networks were the detectors, with a threshold detector as the baseline.

The finding that mattered most here is structural. Window size and sampling rate cannot be chosen independently. The number of samples in a window is their product, and the time a detector waits before it can classify anything equals the window length. A window length chosen without knowing the achieved sampling rate does not specify much. Early drafts of this project treated the two as separate parameters. They are now handled together, and measuring the Arduino’s real per-channel rate with all four channels running is a separate build task instead of a number copied from a datasheet.

A Nyquist argument follows from the same point. A channel sampled at rate f_s can only represent content below $f_s/2$. Frictional vibration during slip has content above what a slow channel can capture, so a detector on a slow channel sees a truncated or aliased signal no matter how good the classifier is. Cravetz et al. (2025) make this point about their own 40 Hz inertial channel, which could not detect frequencies above 20 Hz. The same argument applies to this project, which is why the piezo front end is checked with a 20–200 Hz sweep before any data collection.

Two further pieces were reused. Taking the lag-one difference of the ℓ_2 -norm across channels turns a multi-channel magnitude into a change signal. This removes constant offsets that differ between sensor units and keeps the transients that carry slip information. And because the threshold baseline was competitive in that study, the rule-based detector is kept here as a real experimental condition. If it performs about as well as the random forest, that is worth reporting, because it runs on the microcontroller with no inference cost.

3.3 Dong, Yuan, and Adelson (2017): slip begins at the edges

Using an improved GelSight sensor, Dong et al. (2017) tracked markers printed on the gel and found that relative marker motion appears first at the edge of the contact region and moves inward as the sliding area grows. Slip is a spatial process with a measurable start, not a single instant.

This project has no imaging sensor, so the finding matters mostly for labeling. If slip starts at the contact edges before the object moves, the vibration a piezo disc picks up during incipient slip will arrive before any position-based reference can mark onset. Labeling a window as slip only once the video shows displacement would throw away the earliest part of the signal, which is also the most useful part. Training windows are therefore anchored at the reference onset and extended backward so the pre-onset interval falls inside the labeled region.

Because the GelSight membrane is the fingertip, every trial in that study used the same contact surface, and fingertip materials could not be compared. Surface condition is a variable in this project.

Table 2. Sources and the design decisions they produced. Sources that did not change the design are omitted.

Source	Finding used	Resulting decision
Romano et al. (2011)	Pressure at 24.4 Hz and acceleration at 3 kHz on one robot; proportional recovery; no vision in the loop	Two sensors at different bandwidths; adaptive controller kept as a Pillar Three condition; vision hands off at contact
Van Wyk & Falco (2018)	Detection time equals window length; samples per window = window $\times$ rate; lag-one difference of ℓ_2 -norm; threshold baseline competitive	Measure sampling rate before fixing windows; test 25/50/100 ms against the budget; keep the threshold detector as a real condition
Dong, Yuan & Adelson (2017)	Slip spreads inward from the contact edge before the object moves	Label windows backward from reference onset so the pre-onset interval is included
Cravetz et al. (2025)	195 trials with slip and no-slip; labels from motion capture; network generalized across grippers; 40 Hz IMU limit	Labels from tracked position with a numerical criterion; 30 no-slip trials; single-gripper scope listed as a limitation
Calandra et al. (2018)	Touch improves grasp-outcome prediction over vision alone	Supports vision for initialization and touch for correction
Li, Dong & Adelson (2018)	88.03% slip classification on 10 unseen objects over 152 grasps with optical tactile data	Reference point from an expensive instrument for positioning the low-cost result

3.4 Cravetz et al. (2025): automatic labeling and inertial sensing

Cravetz et al. (2025) tested whether fingertip orientation change and slip-induced vibration, measured by an IMU on a tendon-driven compliant hand, can indicate slip. They ran 195 manipulation trials with slip and no-slip conditions, labeled the data automatically from motion capture, and trained a convolutional network on it. Slip was detected from the inertial data even with other disturbances present, and the trained network still worked on data from a different gripper holding an object it had not seen.

Three things follow for this project. First, automatic labeling is the model for the labeling method here. Hand-marking slip onset from video is slow, and it lets the researcher decide the ground truth for the main metric. Deriving onset from tracked position against a fixed numerical criterion avoids that. Second, a substantial no-slip set is required. The 30 no-slip trials (stable holds, subthreshold loads, jaw movement, light taps, and start–stop events) are there because a detector that has never seen a harmless disturbance will fire on one. Third, their cross-gripper result is a standard a single-gripper study cannot meet, and Section 6.2 lists this as a limitation.

3.5 Class imbalance

Several of the method sources note that slip datasets are heavily imbalanced. Slip takes up a small part of each trial, so windows drawn from continuous logging are mostly negative. A classifier can score high accuracy by predicting “no slip” almost everywhere. For that reason accuracy is not reported. Precision, recall, F1, and false trigger rate are reported instead, and the trial design aims to produce enough positive windows to train on without synthetic oversampling.

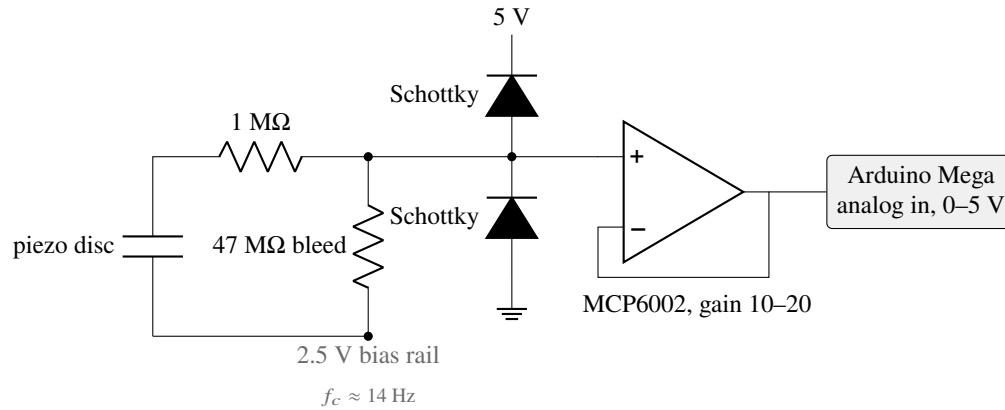


Figure 3. Piezo front end for one channel (feedback network for gain simplified). The bleed resistor gives the disc a DC return path and sets the high-pass corner near 14 Hz. The Schottky pair clamps impact transients to the supply rails. The MCP6002 buffers a node the Arduino input cannot drive directly.

4 System design

4.1 Sensor placement

One FSR sits under each jaw pad, directly behind the contact face, so it measures normal load at the interface and not force carried through the structure. One piezo disc is bonded to each finger body behind the pad. The disc responds to strain in whatever it is bonded to, so this position keeps it out of the load path while still coupling it to vibration traveling through the finger. Sensor positions are recorded at installation, so a failed unit can be replaced in the same spot; moving a sensor partway through the study would quietly change the experiment.

Sensors are installed after the jaw travel check, not during assembly. If the jaws rub anywhere along their travel, fixing it means removing a bonded FSR, and the installation would have to be done twice.

4.2 Piezo signal conditioning

A piezo disc is a small charge source with very high output impedance and no DC path, so it cannot connect straight to a microcontroller analog input. The front end (Figure 3) handles three problems at once. The signal is bipolar and the Arduino input is not, so the channel is biased to 2.5 V and swings around that point. The disc has no DC return, so a 47 MΩ bleed resistor to the bias rail provides one; with the disc's capacitance this sets a high-pass corner near 14 Hz, which removes drift and leaves the frictional band intact. Under impact the disc can produce voltages well outside the supply rails, so a series resistor and a pair of Schottky diodes clamp the node to the rails. An MCP6002 then buffers the high-impedance node and applies a gain of 10 to 20.

Before data collection, a 20–200 Hz sweep is driven into each disc to confirm that the 60–70 Hz band passes through the finished circuit. A high-pass corner slightly in the wrong place would attenuate the content the detector depends on.

4.3 Sampling rate is measured

The target is about 1000 Hz per channel on four channels. Whether an Arduino Mega reaches that while multiplexing four channels, timestamping, and writing serial output is an empirical question, and the answer

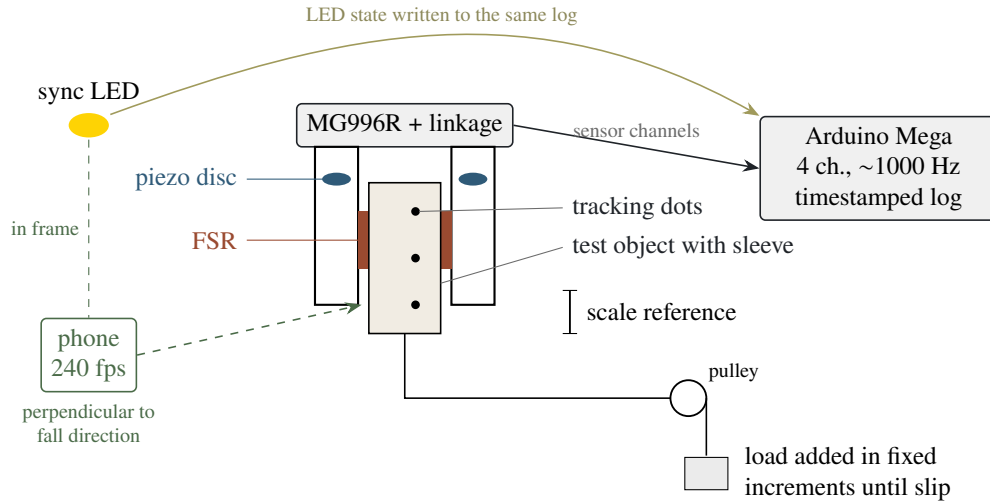


Figure 4. Test apparatus. The object is held between the jaws with FSRs at the contact faces and piezo discs on the finger bodies. Load is added through a cord and pulley in fixed increments until the object slips. The camera views the object perpendicular to the fall direction. One LED flash appears in both the video and the serial log.

changes what each window length means. Measuring the achieved rate with the full acquisition running is therefore its own build task, done before FSR calibration and before the piezo sweep. The measured value will be reported whether or not it meets the target.

It helps to decide in advance what a shortfall would mean. At 1000 Hz a 25 ms window holds about 25 samples per channel, which is enough for a usable band-energy estimate. If the real rate is much lower, the shortest window may not hold enough samples for the feature set. In that case the minimum usable window length is set by the hardware, and it will be reported that way.

4.4 Reference labels for slip onset

Slip onset comes from video, not from the sensors under test; using a sensor to label its own training data would make the evaluation circular. A phone records at 240 fps, mounted perpendicular to the fall direction, with tracking dots on the object and a scale reference in frame (Figure 4). Object position is extracted frame by frame in Vernier Video Analysis.

The onset criterion is numerical and fixed before collection starts. First, a ten-second clip of a stationary grasped object is recorded, and the standard deviation σ of the measured position over that clip gives the noise floor of the method. Slip onset is the first of three consecutive frames whose displacement exceeds 3σ . The stationary clip is also run through the same criterion as a control: no three consecutive stationary frames may cross it. If they do, the criterion or the camera setup is fixed before any trial runs.

The LED joins the two clocks. Its switch-on is written into the serial log and its first lit frame is visible in the video, so both records share a zero. Each trial’s video and serial files are aligned on that frame during analysis. At 240 fps one frame is about 4.2 ms, which is the resolution limit on any reported detection delay.

4.5 Window lengths and the response budget

Three window lengths are tested: 25, 50, and 100 ms. They are three points on one trade-off. A longer window has more samples and finer frequency resolution, which should help the classifier. It also cannot

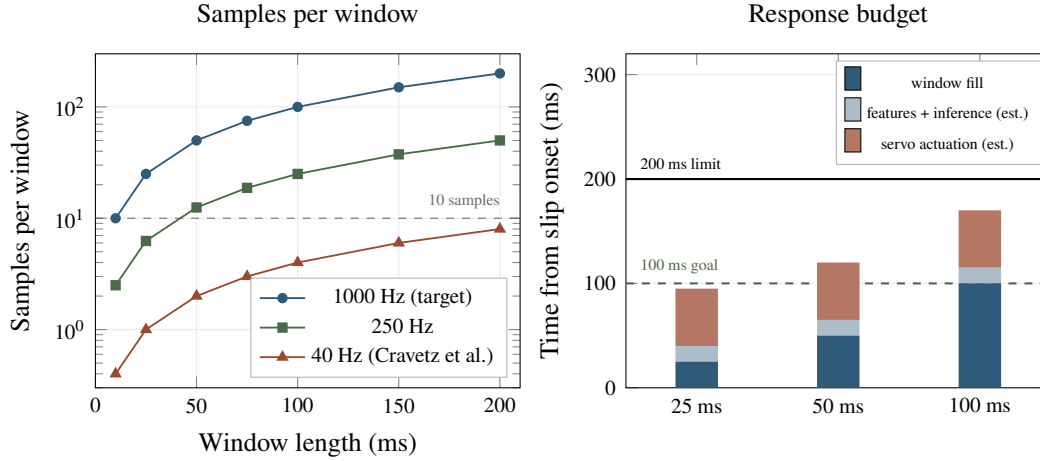


Figure 5. Left: samples inside a window versus window length at three sampling rates. The relationship is multiplicative, so the achieved rate must be known before a window length can be interpreted. Right: how each window length spends the response budget. Window fill is fixed by definition. The feature, inference, and actuation times are planning estimates that will be replaced by measured values.

Table 3. Features, computed per channel at each window length. Band energy is the feature most sensitive to a sampling-rate shortfall, since a short window at a low rate may not hold enough samples to estimate it.

Feature	What it captures	Most relevant channel
Mean	Load level within the window	FSR
Standard deviation	Activity independent of level	Piezo
Slope	Direction and rate of load transfer	FSR
Range	Peak-to-peak excursion; sensitive to single transients	Piezo
RMS	Energy in the window regardless of sign	Piezo
Band energy	Energy in the frictional vibration band	Piezo

return an answer until it has filled, and that time comes out of the response budget before any computation starts.

The response budget is 200 ms from slip onset to servo correction, with 100 ms as a stretch goal. A 100 ms window uses up the whole stretch goal just filling, before feature extraction, inference, or servo motion (Figure 5, right). It may still meet the 200 ms limit, but it carries a known handicap that has to be reported next to its accuracy. Ranking detectors on classification scores alone, without the delay each one costs, would be misleading, so every detector result is reported with its detection delay.

4.6 Features

Features (Table 3) are computed per channel within each window and concatenated across the channels in a given sensing condition. Normalization and filter parameters are fit on training data only and applied unchanged to validation data, so no information from the evaluation set leaks into preprocessing.

Table 4. Variables.

Type	Variable	Levels
Independent	Sensing condition	FSR only; piezo only; fused
Independent	Detector	Rule-based threshold; random forest
Independent	Window length	25, 50, 100 ms
Independent	Surface sleeve	Three levels, low to high friction
Independent	Object mass	100, 250, 400 g
Dependent	Detection quality	Precision, recall, F1, false trigger rate
Dependent	Detection delay	ms from reference onset to detector output
Controlled	Initial jaw opening	From measured width via a frozen lookup table
Controlled	Loading protocol	Fixed increment, wait interval, and start position
Controlled	Sensor positions	Recorded at installation, reproduced on replacement
Controlled	Supply	Regulated 6 V rail, common ground with the Arduino

4.7 Detectors and validation

Two detectors are compared. The rule-based detector fires when the piezo channel rises above one threshold while the FSR channel falls below another; both thresholds are set from pilot data by a stated rule and then frozen. The random forest (scikit-learn) is trained on the full feature vector. Both are scored on the same windows under the same protocol.

Validation uses five-fold cross-validation grouped by trial, so all windows from one trial land in the same fold. Without grouping, windows from a single trial would appear in both training and validation sets. Consecutive windows from one trial are strongly correlated, so the scores would be inflated. Of all the ways this study could report a wrong number that looks right, this is the most likely one, and grouping is treated as a requirement.

Three sensing conditions, two detectors, and three window lengths give 18 cells, each scored on the same five folds. This is what makes RQ1 answerable with evidence instead of a few examples.

5 Experimental plan

5.1 Variables

Table 4 lists the variables. Sensing condition, detector, and window length are crossed in analysis. Surface sleeve and object mass are crossed in data collection.

5.2 Trial matrix

Crossing three surface sleeves with three object masses gives nine conditions (Figure 6). Each condition gets five pilot trials, 45 in total. The pilots set the parameters the final trials depend on: loading increment, wait interval between increments, safe drop distance, correction increment, and the FSR and servo safety limits. These values are locked after the pilot phase. The lookup table from object width to jaw opening is built from pilot data and frozen at the same time.

The final set is 90 slip trials, ten per condition, run in randomized order with video and serial files saved under matching trial numbers. Another 30 no-slip trials are recorded outside the grid: stable holds, subthreshold loads, deliberate jaw movement, light taps, and start–stop events. These negatives are what

surface sleeve	sleeve 1 low friction	10 slip trials + 5 pilot	10 slip trials + 5 pilot	10 slip trials + 5 pilot	30 no-slip trials recorded out- side the grid: stable holds, subthreshold loads, jaw movement, light taps, start-stop
	sleeve 2 medium	10 slip trials + 5 pilot	10 slip trials + 5 pilot	10 slip trials + 5 pilot	
	sleeve 3 high friction	10 slip trials + 5 pilot	10 slip trials + 5 pilot	10 slip trials + 5 pilot	
		100 g	250 g	400 g	
		object mass			

Figure 6. Trial matrix: 9 conditions, 45 pilot trials, 90 slip trials, and 30 no-slip trials held outside the grid. Sleeves are described by friction level because the specific materials are chosen during the pilot phase.

make the false trigger rate a real measurement.

5.3 Procedure for one slip trial

1. Fit the selected sleeve and mass insert. Place the object at the marked start position.
2. Command the jaw opening from the lookup table for the measured width, then close to the fixed initial grip command.
3. Start the serial log and the video, then flash the sync LED once.
4. Add load through the cord and pulley in the locked increment, waiting the locked interval between increments, until the object slips.
5. Stop recording once the object has fallen the safe drop distance. Save both files under the same trial number.
6. Extract object position in Vernier Video Analysis and apply the 3σ , three-frame criterion to mark reference onset.

5.4 Metrics

Table 5 lists the reported metrics. Accuracy is left out because the windowed dataset is dominated by negative windows.

5.5 Handoff to Pillar Three

Pillar Two ends with a chosen detector and window length. Pillar Three puts that detector inside two controllers and runs 140 trials on five unfamiliar objects plus crushable cups. One controller holds a fixed command and only logs detector output; the other tightens by a fixed increment after a confirmed slip and waits one interval before reconsidering. The detection delay measured in Pillar Two feeds the total response-time calculation there. Nothing else crosses the boundary, so the detector evaluation does not depend on whether the recovery controller works.

Table 5. Reported metrics.

Metric	Definition	Reason
Precision	Share of detector firings that match real slip	A detector that fires constantly is useless even with perfect recall
Recall	Share of real slip events detected	A missed slip is a dropped object
F1	Harmonic mean of precision and recall	One comparable number across 18 cells
False trigger rate	Firings per minute during no-slip trials	A false positive means an unnecessary squeeze on a fragile object
Detection delay	ms from reference onset to detector output	Decides whether the 200 ms budget can be met
Achieved sampling rate	Measured per-channel rate with four channels running	Sets what each window length actually means

6 Risks

Table 6 lists the failure modes the design anticipates, each with a response decided in advance. Deciding the response afterward would be hard to distinguish from changing the experiment to fit the result. The third row matters most: it turns the most likely way the project could fail into a result that can still be reported.

7 Novelty and limitations

7.1 Claim

The claim is deliberately narrow. Slip detection, sensor fusion, FSRs, and piezo discs are all established. What my search did not find is a study that pairs these two low-cost sensors on the same gripper, runs a lightweight classifier on the combination, and quantifies how the result compares with the optical and structured tactile sensors behind most published work. The contribution is a measurement and a comparison.

If an earlier study with the same sensor pair and a comparable evaluation turns up, the novelty claim fails and this becomes a replication. That would still be worth writing up, but it would have to be described as a replication. The search behind the claim is documented in the annotated bibliographies and will be repeated before final submission.

7.2 Limitations

- *One gripper.* All trials use one 3D-printed parallel-jaw gripper. Cravetz et al. (2025) showed generalization across grippers; this study cannot, and its results apply to this hardware until someone tests otherwise.
- *Rigid objects.* The test object is rigid with interchangeable sleeves. Deformable objects bring contact mechanics this design does not cover.
- *Induced slip.* Slip is produced by adding load through a pulley in fixed steps. This is repeatable but differs from slip during a real manipulation.
- *Frame-rate resolution.* At 240 fps one frame is about 4.2 ms, which limits how precisely detection delay can be reported.
- *Offline evaluation.* Detectors are trained and scored offline on logged windows. Inference time on the

Table 6. Risk register.

Risk	Why it could happen	Planned response
Sampling rate well below 1000 Hz	Four multiplexed channels, timestamps, and serial output share one microcontroller	Measure first; report the measured rate; if 25 ms cannot support band energy, report a hardware-imposed minimum window
Servo noise buries the piezo signal	MG996R shares the supply and the structure with the sensors	Regulated rail with common ground; stationary noise recording before trials; servo-only recordings kept as a noise reference
Incipient-slip signal too weak to classify	Low-cost sensors may not resolve the pre-onset interval	Narrow the question to signal characterization: report signal-to-noise per condition and the detectability floor
Unreliable reference labels	Camera angle, lighting, or dot tracking degrade position extraction	Check the criterion on a stationary clip first; no three consecutive stationary frames may cross it
Inflated cross-validation scores	Consecutive windows from one trial are strongly correlated	Group folds by trial; state the grouping in the results as well as the methods
Random forest wins against a poorly tuned baseline	A weak baseline flatters the model	Set thresholds from pilot data by a stated rule, freeze them, and report the rule
Sensor drift or unit variation	FSR response differs between units and drifts with repeated loading	Calibrate against known masses, five readings per mass; record positions for replacements

microcontroller is measured separately in Pillar Three; an offline F1 score does not show whether the detector runs fast enough.

- *Sample size.* Ten slip trials per condition can reveal a large effect but not a small one. Confidence intervals are reported alongside point estimates.

8 Status and next steps

The completed work is design and literature review. The hardware list is fixed, the conditioning circuit is designed, the labeling method has a numerical criterion, the trial matrix is written, and the analysis protocol, including fold grouping, is set in advance. The design decisions in Section 4 each trace back to a source in Section 3.

Nothing physical exists yet. The next steps, in order: finish the CAD and print the gripper body and jaws; assemble the servo and linkage; confirm the jaws close evenly across their full travel; install the sensors and record their positions; build and validate the piezo front end; and measure the achieved per-channel sampling rate with all four channels running. That last measurement gates everything after it, because no window length in this report has a fixed meaning until it is known. Figure 7 shows the phase plan.

References

Calandra, R., Owens, A., Jayaraman, D., Lin, J., Yuan, W., Malik, J., Adelson, E. H., & Levine, S. (2018). More than a feeling: Learning to grasp and regrasp using vision and touch. *IEEE Robotics and Automation*

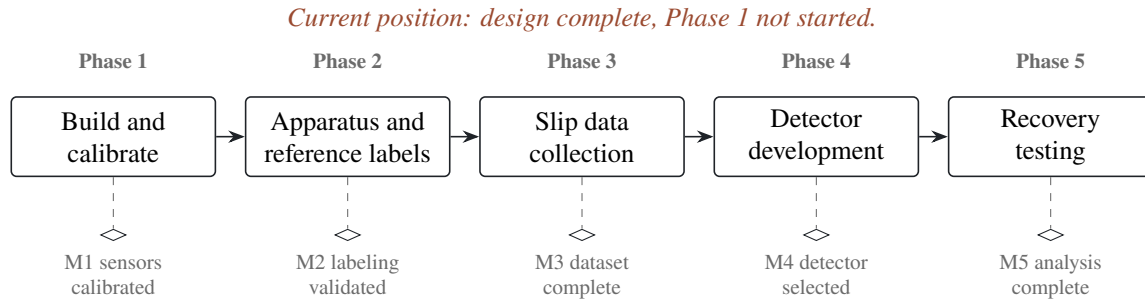


Figure 7. Phase plan and milestones. Phases 1 and 2 can largely run in parallel, since the test rig, loading setup, and camera do not depend on the gripper. They first meet at the stationary noise recording, which needs a working grip and a working camera at the same time.

Letters, 3(4), 3300–3307.

Cravetz, M., Vyas, P., Grimm, C., & Davidson, J. R. (2025). Slip detection for compliant robotic hands using inertial signals and deep learning. *Frontiers in Robotics and AI*, 12. <https://doi.org/10.3389/frobt.2025.1698591>

Dong, S., Yuan, W., & Adelson, E. H. (2017). Improved GelSight tactile sensor for measuring geometry and slip. In *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 137–144).

Li, J., Dong, S., & Adelson, E. H. (2018). Slip detection with combined tactile and visual information. In *2018 IEEE International Conference on Robotics and Automation (ICRA)* (pp. 7772–7777).

Romano, J. M., Hsiao, K., Niemeyer, G., Chitta, S., & Kuchenbecker, K. J. (2011). Human-inspired robotic grasp control with tactile sensing. *IEEE Transactions on Robotics*, 27(6), 1067–1079.

Van Wyk, K., & Falco, J. (2018). Calibration and analysis of tactile sensors as slip detectors. In *2018 IEEE International Conference on Robotics and Automation (ICRA)*. <https://doi.org/10.1109/ICRA.2018.8461117>